

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 09.1

Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation.

Name: Dhinakaran. S **Reg. No.:** 192412353

Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;

% Input and echo path
x = randn(1,1000);
h = [0.8 0.5 0.3];
d = filter(h,1,x);

% LMS parameters
mu = 0.01;
M = length(h);
w_lms = zeros(1,M);
y_lms = zeros(1,length(x));
e_lms = zeros(1,length(x));

% NLMS parameters
mu_n = 0.8;
w_nlms = zeros(1,M);
y_nlms = zeros(1,length(x));
e_nlms = zeros(1,length(x));

for n = M:length(x)
    u = x(n:-1:n-M+1);
    y_lms(n) = w_lms*u.';
    e_lms(n) = d(n)-y_lms(n);
    w_lms = w_lms + mu*e_lms(n)*u;

    y_nlms(n) = w_nlms*u.';
    e_nlms(n) = d(n)-y_nlms(n);
    w_nlms = w_nlms + (mu_n/(u*u.'+1e-8))*e_nlms(n)*u;
end

figure; plot(x); hold on; plot(d); grid on;
legend('Input','Echo'); title('Input and Echo Signal');
figure; plot(e_lms); grid on; title('LMS Error / Echo-Cancelled Output');
figure; plot(e_nlms); grid on; title('NLMS Error / Echo-Cancelled Output');
figure; semilogy(e_lms.^2+eps); hold on; semilogy(e_nlms.^2+eps);
grid on; legend('LMS','NLMS'); title('MSE Comparison');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
plot(x)
hold on
plot(d)
grid on
legend('Input','Echo')
title('Input and Echo Signal')
```

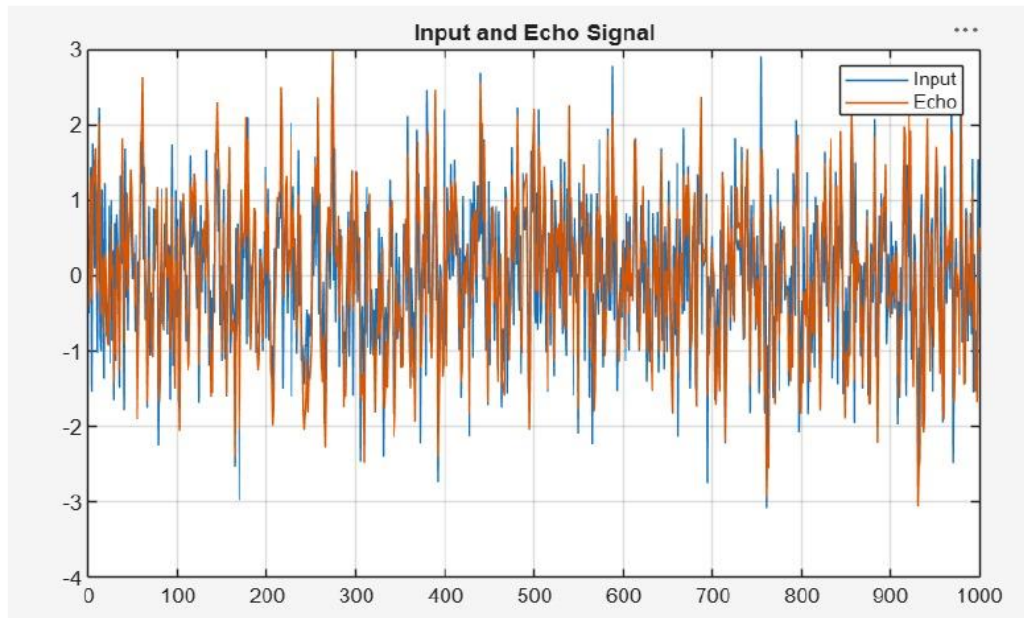
```
figure
plot(e1)
grid on
title('LMS Output')
```

```
figure
plot(e2)
grid on
title('NLMS Output')
```

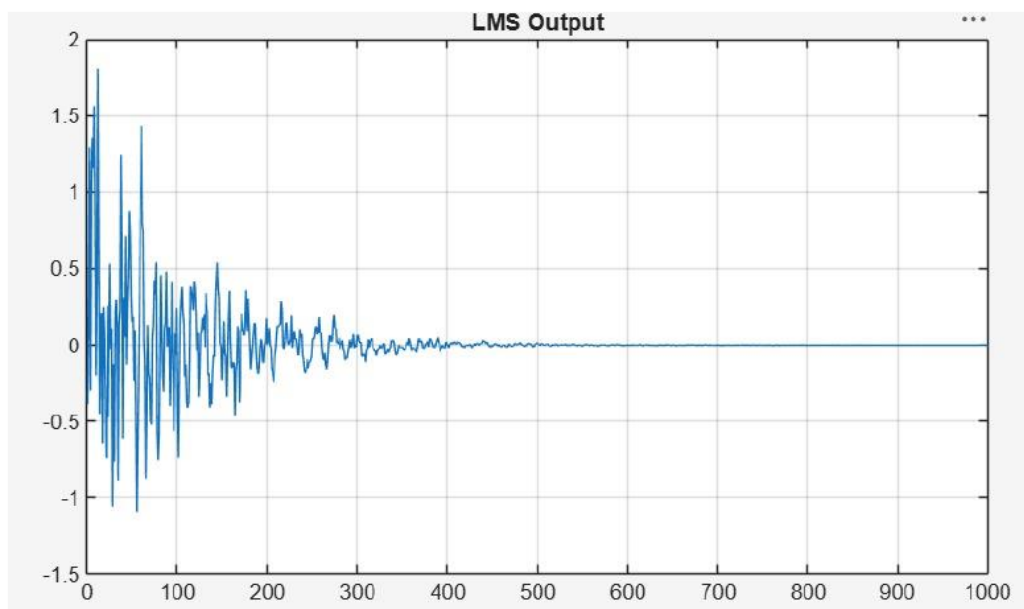
```
figure
semilogy(e1.^2,'r')
hold on
semilogy(e2.^2,'b')
grid on
legend('LMS','NLMS')
title('MSE Comparison')
```

Output:

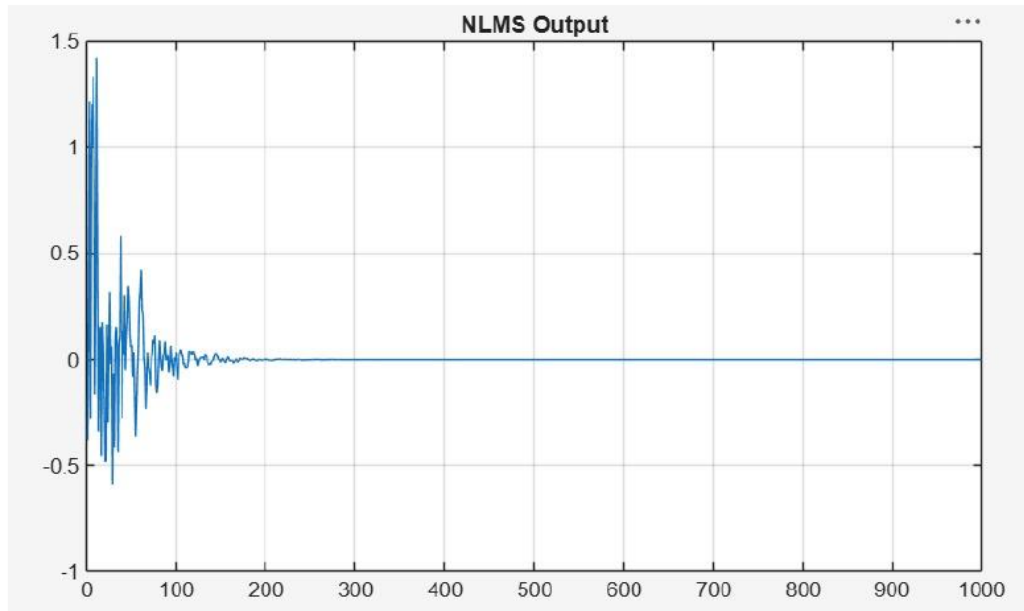
Objective 1:



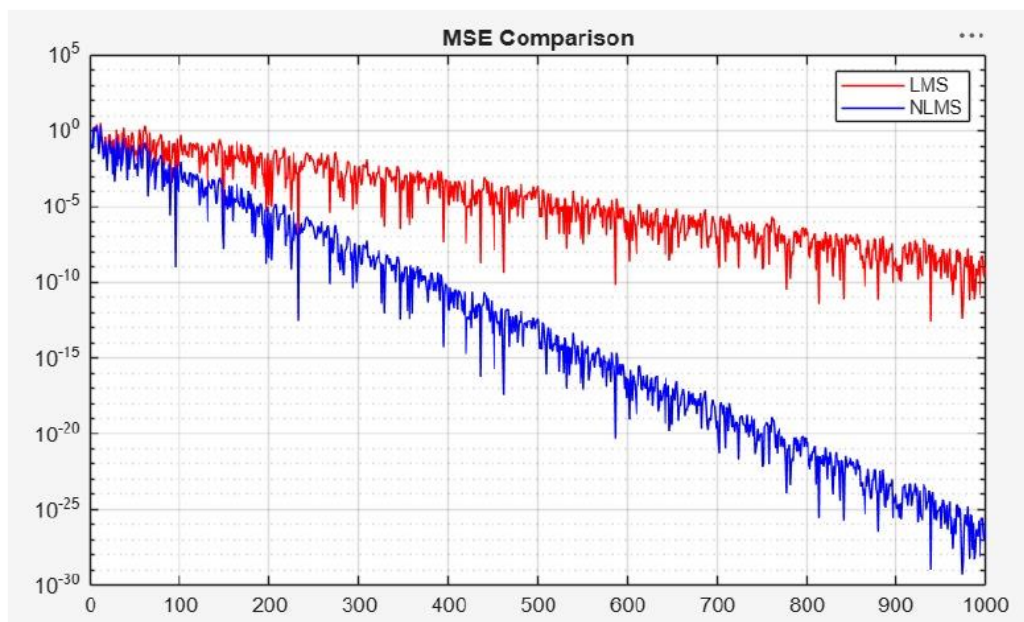
Objective 2:



Objective 3:



Objective 4:



Result:

1. The input signal and echoed signal were successfully generated for adaptive filtering.
2. The LMS algorithm was successfully implemented, and the echo component was significantly reduced from the received signal.
3. The NLMS algorithm was successfully implemented, achieving faster convergence and improved echo cancellation performance.
4. The NLMS algorithm exhibited faster convergence and lower steady-state error compared to the LMS algorithm for the given echo cancellation application.

Practical Question 1:

An input signal of 5000 samples is applied to an echo path modeled as a 16-tap FIR filter. Design and implement adaptive echo cancellers using both LMS and NLMS algorithms in MATLAB to remove the echo component from the received signal. Compare the performance of the two algorithms based on convergence behavior and mean square error.

Tasks

1. Generate an input signal of 5000 samples and simulate an echo path using a 16-tap FIR filter for adaptive echo cancellation.
2. Implement the LMS adaptive filtering algorithm in MATLAB to estimate and cancel the echo component from the received signal.
3. Implement the NLMS adaptive filtering algorithm in MATLAB to estimate and cancel the echo component from the received signal.
4. Compare the performance of the LMS and NLMS algorithms by analyzing their convergence characteristics and Mean Square Error (MSE).

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2:

An input signal of 8000 samples is applied to an echo path modeled acoustic echo represented by a 32-tap FIR filter. Using MATLAB, implement LMS and NLMS adaptive filters to estimate and cancel the echo. Evaluate the effectiveness of both algorithms under identical operating conditions.-give task similar to objective

Tasks

1. Generate an input signal of 8000 samples and model the acoustic echo path using a 32-tap FIR filter.
2. Implement the LMS adaptive filtering algorithm in MATLAB to estimate the echo path and perform echo cancellation.
3. Implement the NLMS adaptive filtering algorithm in MATLAB to estimate the echo path and perform echo cancellation.

4. Evaluate and compare the performance of the LMS and NLMS algorithms using convergence characteristics and Mean Square Error (MSE) under identical operating conditions.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To implement LMS and NLMS echo cancellation and compare convergence and MSE. The executable version above preserves the manual intent while replacing the malformed placeholder exponential lines with actual adaptive updates.

Student Details

Name: Dhinakaran. S

Reg. No.: 192412353

Course: ECA09 - Digital Signal Processing